

APPENDIX A: MATHEMATICAL FORMULATIONS & QUANTUM CIRCUIT ENCODINGS

A.1 Quantum State Preparation & AngleEmbedding Ansatz

Classical input vectors $\mathbf{x} \in \mathbb{R}^D$ (where $D=2$ in the matched experiment: TransactionAmt and card1) are normalized to the interval $[0, \pi]$. The state preparation unitary $U(\mathbf{x})$ maps classical features into product states via single-qubit rotations followed by entangling gates:

$$|\phi(\mathbf{x})\rangle = U_{\text{ent}} \cdot \bigotimes_{i=1}^D R_Y(x_i) |0\rangle^{\otimes D}$$

where $R_Y(\theta) = \exp(-i \theta Y / 2)$ and U_{ent} applies alternating CNOT operations across adjacent qubits. For two qubits, this yields the entangled statevector:

$$|\psi(\mathbf{x})\rangle = \cos(x_1/2)\cos(x_2/2)|00\rangle + \cos(x_1/2)\sin(x_2/2)|01\rangle + \sin(x_1/2)\sin(x_2/2)|10\rangle + \sin(x_1/2)\cos(x_2/2)|11\rangle$$

A.2 Projected Quantum Kernel (PQK) Formulation

Standard fidelity quantum kernels evaluate $k_F(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2$. As Hilbert space dimensionality grows, fidelity kernels suffer from exponential concentration of measure (the quantum vanishing gradient / barren plateau phenomenon for kernels), causing kernel values to concentrate around zero.

To mitigate concentration, we implement a **Projected Quantum Kernel (PQK)** following Huang et al. (2021). The quantum state is projected back to a classical feature space by evaluating local one-qubit observable expectations:

$$f_i(\mathbf{x}) = \langle \phi(\mathbf{x}) | \sigma_z^{(i)} | \phi(\mathbf{x}) \rangle, \quad \text{for } i \in \{1, \dots, D\}$$

The resulting projected vector $\mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), \dots, f_D(\mathbf{x})]^T$ is then evaluated via a Gaussian radial basis function:

$$k_{\text{PQK}}(\mathbf{x}, \mathbf{x}') = \exp(-\gamma_P \cdot \|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\|^2)$$

This projection preserves non-linear quantum correlations while retaining non-trivial kernel contrast across large transaction batches. Note that our Amazon Braket evaluation uses $\text{shots}=0$ (exact statevector analytic expectation values) rather than shot-sampling, ensuring the latency reported represents pure execution overhead without sampling noise.

A.3 Centered Kernel Alignment (CKA) Metric Space Geometry

To evaluate whether the quantum kernel explores an orthogonal metric space relative to classical kernels, we compute the Centered Kernel Alignment (Cortes et al., 2012). Let K_1 and K_2 be the Gram matrices of the quantum and classical kernels on the identical $N=200$ support set. The centered Gram matrix is $K_c = H K H$, where $H = I - (1/N) \mathbf{1} \mathbf{1}^T$ is the centering matrix. CKA is defined as:

$$\text{CKA}(K_1, K_2) = \text{Tr}(K_{c1} K_{c2}) / \sqrt{\text{Tr}(K_{c1}^2) \cdot \text{Tr}(K_{c2}^2)} \in [0, 1]$$

Our measured CKA of **0.5741** between PQK and Classical RBF demonstrates that the quantum kernel metric geometry is highly collinear with the classical Gaussian kernel, explaining why the empirical decision boundary showed no statistically significant separation.

A.4 Matrix Product State (MPS) Simulability Diagnostic

For any quantum circuit proposed for payment fraud detection, classical simulability is governed by bipartite entanglement entropy:

$$S(A:B) = -\text{Tr}(\rho_A \cdot \log_2 \rho_A)$$

If $S(A:B) \leq c \cdot \log(N)$, the circuit can be simulated in polynomial time on classical workstations via Matrix Product States (MPS) with bounded bond dimension χ . Under our 2-qubit to 12-qubit architectures, $\chi \leq 64$ suffices for exact representation, confirming that classical hardware remains capable of simulating these feature maps without requiring physical QPUs.

APPENDIX B: GRANULAR EXPERIMENTAL EVIDENCE & LATENCY PROFILES

B.1 Complete Operational Review Budget Sweep

To establish the complete trade-off curve between manual review capacity and fraud interception, the uncertainty router was evaluated across five operational budgets on the out-of-sample forward test set (118,108 transactions, 4,064 total frauds):

Review Budget (B)	Escalated Volume (Tx)	Frauds Captured (N)	Fraud Density (%)	Prevalence Enrichment	Cumulative Stream Recall	Amount Policy Frauds	Ablation Advantage
0.5%	591 tx	253	42.81%	12.44x	6.23%	9 frauds (1.52%)	28.1x more fraud
1.0%	1,181 tx	440	37.26%	10.83x	10.83%	34 frauds (2.88%)	12.9x more fraud
2.0%	2,362 tx	720	30.48%	8.86x	17.72%	152 frauds (6.44%)	4.7x more fraud
5.0%	5,905 tx	1,559	26.40%	7.67x	38.36%	312 frauds (5.28%)	5.0x more fraud
10.0%	11,811 tx	2,343	19.84%	5.77x	57.65%	570 frauds (4.83%)	4.1x more fraud

A spot-check of SHAP top-3 features for correctly escalated frauds confirms they vary significantly per-row (e.g., [card3, C1, C5] vs [C13, card3, C5] vs [C5, C2, C1]), demonstrating the frontline LightGBM relies on diverse transaction signatures rather than a static global ranking.

B.2 End-to-End Latency Breakdown Across Architectural Paths

Benchmarking was conducted on dedicated x86_64 server hardware across 1,000 real benchmark transactions:

Execution Stage / Component	Median Latency	p95 Latency	Max Latency	Design Budget Compliance (≤50 ms)	Hardware Architecture
Raw Ingress & Feature Preprocessing	3.85 ms	4.68 ms	6.12 ms	Fully Compliant (Consumes 7.7% of budget)	Single x86_64 core
LightGBM Scoring + Calibration	0.21 ms	0.28 ms	0.45 ms	Fully Compliant (Consumes 0.4% of budget)	Single x86_64 core
Uncertainty Routing Logic	0.01 ms	0.01 ms	0.02 ms	Fully Compliant (<0.1% of budget)	Single x86_64 core
Total Autonomous Fast Path	4.07 ms	4.97 ms	6.59 ms	Fully Compliant (Consumes <10% of budget)	Commodity CPU
Escalated Classical Specialist (Tuned RBF)	0.82 ms	0.48 ms	1.15 ms	Fully Compliant (Total latency: 4.89 ms)	Commodity CPU
Local Quantum Simulator (PennyLane N=200)	159.14 ms	217.18 ms	342.50 ms	Non-Compliant (3.2x – 4.3x over 50 ms budget)	Local CPU Statevector
Cloud QPU Remote Queue (IonQ Aria / Braket)	180,000 ms (180 s)	1,200,000 ms (20 min)	3,600,000 ms (1 hr)	Severely Incompatible with Real-Time Clearing	Trapped-Ion Cloud Hardware

B.3 Unit Economics & Break-Even Accounting

ECONOMIC PROVENANCE DECLARATION

Realized Savings to Date: \$0.00. The research uses historical IEEE-CIS data; no live banking operations have accrued financial savings. All economic figures are modeled under explicit cost assumptions (\$0.50/1M frontline LightGBM compute; \$35.30 per escalated transaction on IonQ Aria via AWS Braket rate cards).

At a 1.0% escalation budget over 1,000,000 transactions (10,000 escalated transactions), the selective classical architecture incurs \$0.15 in additional compute, breaking even if it intercepts just 1 additional fraud. In contrast, cloud QPU execution incurs \$353,000 in compute expenses. Under the observed null result (Δ PR-AUC = -0.1021, p = 0.246), QPU execution produces an immediate net financial deficit of over \$167,000.

APPENDIX C: REPRODUCIBILITY MANIFESTO & ACADEMIC REFERENCES

C.1 Reproducibility Protocol & Provenance Tracking

Every empirical metric, figure, and table in this proposal is fully reproducible from the public code repository. The complete experimental harness is frozen under git commit a2aa2d1.

```
# Clone and verify the complete research suite git clone https://github.com/atharveeee-netizen/hsbc-quantum-fraud.git cd
hsbc-quantum-fraud python -m venv .venv && source .venv/bin/activate pip install -r requirements.txt pytest # Executes 24
automated unit, integration, and firewall tests
```

The repository includes: (1) docs/evidence/evidence_ledger.json indexing SHA-256 checksums and parameter manifests for all 16 primary experimental runs; (2) automated claim firewall tests (test_claim_firewall.py) that block unauthorized buzzwords and unverified metrics; and (3) a 7-panel interactive Streamlit dashboard (src/dashboard/app.py) enabling real-time inspection of routing concentration, bootstrap distributions, and latency curves.

The evaluation pipeline uses a fixed random state (random_state=42) for PCA decomposition, scaling, and cross-validation splitting to guarantee bit-identical reproducible metric outputs across executions.

C.2 Canonical References & Prior Literature

1. **Chaves, R., Kumar, K., Chagas, B., Linerud, R., Sorem, B., Mancilla, J., Bell, B. (2026).** A Mixture-of-Experts Framework for Practical Hybrid-Quantum Models in Credit Card Fraud Detection. *arXiv:2603.06473v2*. Demonstrates selective routing to hybrid quantum experts, reporting 7–21 minute added inference latencies.

2. **Mastercard & Oxford Quantum Circuits (OQC) (2026).** Bringing Quantum to Real Payments: A New Approach to Fraud Detection. White paper evaluating hybrid quantum ensembles on payment network authorization flows.

3. **Flórez Ablan, R., Roth, M., Schnabel, J. (2025).** On the similarity of bandwidth-tuned quantum kernels and classical kernels. *Quantum Science and Technology*. Demonstrates that optimal bandwidth tuning causes quantum kernels to converge geometrically to classical Gaussian RBF behavior.

4. **Innan, N., Singh, A., Shafique, M. (2025).** CircuitHunt: Automated Quantum Circuit Screening for Superior Credit-Card Fraud Detection. *IEEE QAI 2025*, pp. 432–438. Reports high accuracy under artificial SMOTE rebalancing.

5. **Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H., McClean, J. R. (2021).** Power of data in quantum machine learning. *Nature Communications*, 12(1), 2631. Establishes the mathematical foundation for Projected Quantum Kernels (PQK) and geometric difference bounds.

6. **Cortes, C., Mohri, M., Rostamizadeh, A. (2012).** Algorithms for learning kernels based on centered alignment. *Journal of Machine Learning Research*, 13, 795–828. Defines the Centered Kernel Alignment (CKA) metric for comparing representation topologies.

7. **Kyriienko, O., Magnusson, E. B. (2022).** Unsupervised quantum machine learning for fraud detection. *arXiv:2208.01201*. Evaluates quantum kernel methods on subsampled, artificially balanced fraud datasets.

8. **IEEE-CIS Fraud Detection Benchmark (2019).** Vesta Corporation & IEEE Computational Intelligence Society. Dataset comprising 590,540 real-world card-not-present e-commerce transactions across 394 features.

C.3 Verification Manifest & Hardware Gate Audit

Artifact Component	Repository Path	Verification Hash / Identifier	Gate Status
Actual Escalated Cohort	data/real/escalated_cohort.npz	N=200 Support & N=200 Eval (8 PCA feats)	Shipped Deliverable
Full Scientific Suite	src/evaluation/run_real_scientific_suite.py	Deterministic execution (Seed: 42)	Verified [MEASURED]
Matched Quantum Experiment	docs/evidence/real_quantum_matched_experiment.json	N=200 Support; 1,000 Paired Resamples	Null Retained (p = 0.246)
Router Policy Ablation	docs/evidence/real_router_ablation.json	5-Budget Sweep vs Amount Baseline	28.1x Lift Confirmed
Hardware Decision Gate	docs/evidence/hardware_gate.json	5-Criterion Operational Protocol	HARDWARE NOT JUSTIFIED
Provenance Manifest	docs/evidence/PROVENANCE_MANIFEST.json	16 Indexed Evidence Artifacts	Git HEAD: a2aa2d1